

Lesson 17: Difference Equations and Z-Transforms

Professor Dave Explains Differential Equations Master Series · Lesson 17

Video: Difference Equations and Z-Transforms Course: Applied Ordinary Differential Equations (ENGR 213)

1. Video Intuition & Pedagogical Framing

Professor Dave explains that while differential equations govern smooth continuous time (analog physics), computers, digital sensors, and financial markets operate in discrete clock ticks: $n = 0, 1, 2, \dots$. The discrete equivalent of a differential equation is a Difference Equation (or recurrence relation). Just as the Laplace transform solves continuous ODEs by mapping $t \rightarrow s$, the Z-Transform solves difference equations by mapping discrete sequences $x[n] \rightarrow z$ using complex power series.

2. Core Theoretical Framework & Rigorous Formulations

1. The Unilateral Z-Transform

$$\mathcal{Z}\{x[n]\} = X(z) = \sum_{n=0}^{\infty} x[n]z^{-n}$$

where $z \in \mathbb{C}$ and $|z| > R$ defines the **Region of Convergence (ROC)**.

Sequence $x[n]$ ($n \geq 0$)	Z-Transform $X(z)$	ROC
1 (Unit step)	$\frac{z}{z-1} = \frac{1}{1-z^{-1}}$	$ z > 1$
a^n (Geometric growth/decay)	$\frac{z}{z-a} = \frac{1}{1-az^{-1}}$	$ z > a $
$\delta[n]$ (Unit sample impulse)	1	All z

2. Time-Delay Shifting Property

$$\mathcal{Z}\{x[n-1]\} = z^{-1}X(z) + x[-1], \quad \mathcal{Z}\{x[n+1]\} = zX(z) - zx[0]$$

3. Fully Solved Whiteboard Problems

Engineering Problem: Solving a First-Order Digital Filter Difference Equation

Problem Statement: Solve the discrete difference equation $y[n] - 0.5y[n-1] = (0.2)^n$ for $n \geq 0$, with initial rest condition $y[-1] = 0$.

Step 1: Apply Z-Transform to Both Sides

$$Y(z) - 0.5z^{-1}Y(z) = \mathcal{Z}\{(0.2)^n\} = \frac{1}{1-0.2z^{-1}}.$$

Step 2: Factor and Isolate $Y(z)$

$$(1 - 0.5z^{-1})Y(z) = \frac{1}{1-0.2z^{-1}} \implies Y(z) = \frac{1}{(1-0.5z^{-1})(1-0.2z^{-1})}.$$

Step 3: Partial Fraction Expansion in z^{-1}

$$\frac{1}{(1-0.5z^{-1})(1-0.2z^{-1})} = \frac{A}{1-0.5z^{-1}} + \frac{B}{1-0.2z^{-1}}. \text{ Using the Heaviside cover-up method: } A = \frac{1}{1-0.2(2)} = \frac{1}{1-0.4} = \frac{1}{0.6} = \frac{5}{3}. B = \frac{1}{1-0.5(5)} = \frac{1}{1-2.5} = -\frac{1}{1.5} = -\frac{2}{3}.$$

Step 4: Assemble Decomposed Z-Domain Response

$$Y(z) = \frac{5/3}{1-0.5z^{-1}} - \frac{2/3}{1-0.2z^{-1}}.$$

Step 5: Inverse Z-Transform Lookup

$$\text{Using } \mathcal{Z}^{-1}\{\frac{1}{1-az^{-1}}\} = a^n: y[n] = \frac{5}{3}(0.5)^n - \frac{2}{3}(0.2)^n \text{ for } n \geq 0.$$

⚠ Common Exam Pitfall & Trapdoor Avoidance

Be careful when choosing between formulations in z versus z^{-1} . In digital signal processing, rational functions are conventionally expanded in negative powers z^{-1} .

4. Key Takeaways & Exam Strategy Formula Card

Core Takeaway & Exam Rule

The Z-Transform is the discrete-time twin of the Laplace transform: continuous poles in the left-half s -plane map inside the unit circle $|z| < 1$.

- Video reference: [Difference Equations and Z-Transforms](#)
- Always verify boundary conditions and check that candidate solutions satisfy the governing differential equation identically.